

PROJECT SPECIFICATION: GROPS V1

1. ROLE & CORE MISSION

You are a Senior Full-Stack Engineer and AI Architect. Build "Grops V1," a web platform that transforms PDF documents into interactive, hierarchical "Neurographs." The goal is to map knowledge with extreme clarity, distinguishing between correlation and causality.

2. TECHNICAL STACK & ARCHITECTURE

- **Frontend:** React (Next.js or Vite), Tailwind CSS.
- **Graph Engine:** React Flow or Cytoscape.js (optimized for non-overlapping hierarchical layouts).
- **Authentication:** Implement a full Auth system (Clerk, Supabase, or Firebase).
- **AI/LLM:** Integration for PDF parsing and Graph Extraction.
- **Database:** Required for user accounts, graph history, and premium waitlist storage.

3. CORE LOGIC & CONSTRAINTS

- **Input Limits:** Up to 3 documents per graph; Max 50 pages per document.
- **Graph Generation Types:**
 1. **Summary Mode:** < 30 nodes (key concepts only).
 2. **Detailed Mode:** Up to 100 nodes (includes nuanced sub-concepts).
- **Logic Rules:** * Distinguish between **Correlation** (bidirectional/simple link) and **Cause-and-Effect** (directional edges).
 - Each node must contain a visual metadata "signal" identifying its source document.
 - The final graph must have clear "Start" and "Finish" nodes.
- **Performance:** Generation must complete within **30 seconds**.
- **Usage Limit:** Max **3 graphs** per user per month.

4. UI/UX SPECIFICATIONS

- **Visual Style:** Hierarchical, highly organized, and linear. Minimalist, premium aesthetic. Avoid node or edge overlapping at all costs.
- **Landing:** Large Drag & Drop area for PDFs.
- **Workspace:**
 - **Main Canvas:** Interactive graph with zoom/pan.
 - **Right Sidebar:** Context panel for node details or AI explanations.
 - **Prompt Bar (Bottom):** Persistent AI command center.
- **History:** Scrollable chronological view in the home page of the user's previous graphs (Private: users only see their own data).
- **Premium Feature:** A "Premium" button (Top-Right) in home page that triggers a pop-up modal to collect Name and Email for a waitlist to be notified when Premium Features

release for just €14.99/mo.

5. ADMIN & MONITORING (Owner Only)

- **Admin Dashboard:** Create a protected route (e.g., /admin) accessible only to a specific Admin UID.
- **Monitoring Features:**
 - **User Activity:** View total number of users and active sessions.
 - **Usage Tracking:** Monitor how many graphs each user has generated against their limit.
 - **Waitlist Management:** View and export the list of names and emails collected from the Premium modal.
 - **System Health:** Display AI generation success rates and average processing times.

6. INTELLIGENT INTERACTION PATTERNS (Prompt Bar Logic)

The Prompt Bar must handle two distinct intents:

A. Contextual Expansion (New Information)

- **Scenario:** User provides a new document or an external consideration.
- **Behavior:** AI evaluates the input and suggests **new nodes and connections** to the existing graph.
- **Workflow:** 1. AI presents the proposed changes in the sidebar.
2. AI explains the logic behind these new connections.
3. User must "Approve" or "Modify" before the graph is permanently updated.
4. Iterate until the user is satisfied.

B. Internal Clarification (Graph Knowledge)

- **Scenario:** User asks a question about existing nodes or connections.
- **Behavior:** AI **highlights** the relevant path/nodes on the graph.
- **Workflow:** AI provides a summarized text answer in the sidebar, providing deep-dive context based on the source PDFs.